

School Management System

~By, U. Roopasri (PFS-ONE-002)

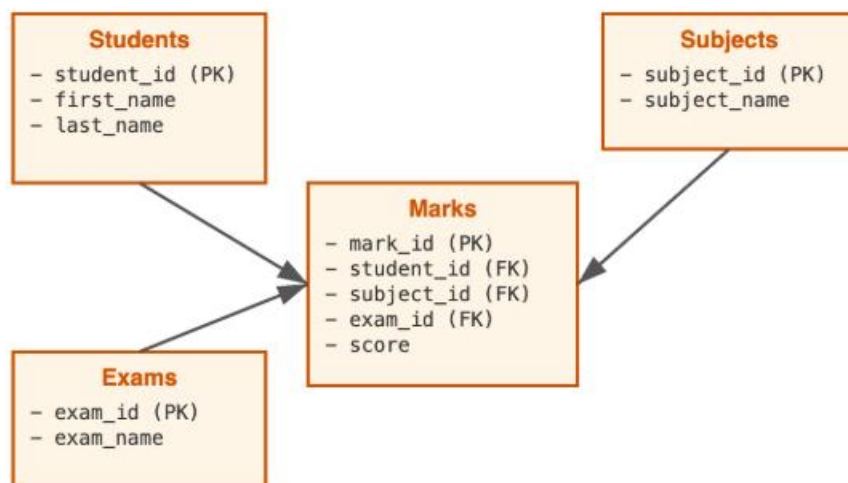
Theme

This Database System manages student information, subjects, exams, and marks while providing easy access in the performance of reports.

Modules

- **Students:** Stores student personal details.
- **Subjects:** Stores the list of subjects (Mathematics, Science, Hindi, etc.).
- **Exams:** Stores details about specific exam events (e.g., Half-Yearly 2024).
- **Marks:** The transaction table linking students, subjects, and exams with a score.

Entity Relationship Diagram (ERD)



```
CREATE DATABASE school_management_system;
```

```
USE school_management_system;
```

Sample Data

Students

Student_id	First_name	Last_name
1	Roopasri	Ummadisetty
2	Bhavani	Puli
3	Priya	Mohan
4	Ananya	Sharma
5	Akshay	Kumar
6	Krunal	Varma
7	Aarav	Varma
8	Vyshnavi	Kumbala
9	Jhansi	Lakshmi
10	Vandana	Rathod

Subjects

Subject_id	Subject_name
1	Mathematics
2	Science
3	Hindi
4	English

Exams

Exam_id	Exam_name
1	Half-yearly 2024
2	Final Exam 2024

Marks

Marks_id	Student_id	Subject_id	Exam_id	Score
1	1	1	1	95
2	1	2	1	88
3	2	1	1	76
4	2	3	1	82
5	3	4	2	90
6	4	2	2	67

Queries

Level 1: Simple (Basic Selection & Filtering)

1.Retrieve all students: List the first name and last name of every student.

```
SELECT first_name, last_name  
FROM Students;
```

	first_name	last_name
▶	Roopasri	Ummadisetty
	Bhavani	Puli
	Priya	Mohan
	Ananya	Sharma
	Akshay	Kumar
	Krunal	Varma
	Aarav	Varma
	Vyshnavi	kumbala
	Jhansi	Lakshmi
	Vandana	Rathod

2.Filter by Score: Find all entries in the Marks table where the score is greater than 80.

```
SELECT * FROM Marks  
WHERE score > 80;
```

	marks_id	student_id	subject_id	exam_id	score
▶	1	1	1	1	95
	2	1	2	1	88
	4	2	3	1	82
	5	3	2	2	90
*	NULL	NULL	NULL	NULL	NULL

3.Sort Data: List all exams sorted by their date, starting with the most recent.

```
SELECT * FROM Exams  
ORDER BY exam_name DESC;
```

	exam_id	exam_name
▶	1	Half-yearly 2024
	2	Final Exam 2024
*	NULL	NULL

Level 2: Intermediate (Joins & Aggregation)

4.Student Marks: List the First Name, Subject Name, and Score for every mark recorded.

```
SELECT s.first_name, sub.subject_name, m.score
FROM Marks m
JOIN Students s ON m.student_id = s.student_id
JOIN Subjects sub ON m.subject_id = sub.subject_id;
```

	first_name	subject_name	score
▶	Roopasri	Mathematics	95
	Bhavani	Mathematics	76
	Roopasri	Science	88
	Ananya	Science	67
	Bhavani	Hindi	82
	Priya	English	90

5. Subject Averages: Calculate the average score for each subject. Display the Subject Name and the Average Score.

```
SELECT sub.subject_name, AVG(m.score) AS average_score
FROM Marks m
JOIN Subjects sub ON m.subject_id = sub.subject_id
GROUP BY sub.subject_name;
```

	subject_name	average_score
▶	Mathematics	85.5000
	Science	77.5000
	Hindi	82.0000
	Englis	90.0000

6.Count Exams: How many marks have been recorded specifically for "Ananya Sharma"?

```
SELECT COUNT(*) AS total_marks
FROM Marks m
JOIN Students s ON m.student_id = s.student_id
WHERE s.first_name = 'Ananya' AND s.last_name = 'Sharma';
```

	total_marks
▶	1

Level 3: Complex (Subqueries, Grouping & Logic)

7.Top Performer: Find the student (First Name, Last Name) who has the highest single score in any exam.

```
SELECT s.first_name, s.last_name, m.score
FROM Marks m
JOIN Students s ON m.student_id = s.student_id
ORDER BY m.score DESC
LIMIT 1;
```

	first_name	last_name	score
▶	Roopasri	Ummadisetty	95

8.Scores by Exam: Display the Exam Name and the total sum of scores for that exam, but only for exams where the total sum is greater than 250.

```
SELECT e.exam_name, SUM(m.score) AS total_score
FROM Marks m
JOIN Exams e ON m.exam_id = e.exam_id
GROUP BY e.exam_name
HAVING SUM(m.score) > 250;
```

	exam_name	total_score
▶	Half-yearly 2024	341

9.Pass/Fail Label: Create a report showing the Student Name, Subject, Score, and a status column that says "Pass" if the score is 70 or above, and "Fail" if below.

```
SELECT s.first_name, s.last_name, sub.subject_name, m.score,
CASE
    WHEN m.score >= 70 THEN 'Pass'
    ELSE 'Fail'
END AS status
FROM Marks m
JOIN Students s ON m.student_id = s.student_id
JOIN Subjects sub ON m.subject_id = sub.subject_id;
```

	first_name	last_name	subject_name	score	status
▶	Roopasri	Ummadisetty	Mathematics	95	Pass
	Bhavani	Puli	Mathematics	76	Pass
	Roopasri	Ummadisetty	Science	88	Pass
	Ananya	Sharma	Science	67	Fail
	Bhavani	Puli	Hindi	82	Pass
	Priya	Mohan	English	90	Pass